

Class 20: Vector Spaces and Subspaces

Definition: A vector space is a nonempty set V of objects called vectors along with two operations, vector addition and scalar multiplication, that satisfy the ten rules listed below for all vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ and all real number scalars c and d .

1. For all $\mathbf{u}$ and $\mathbf{v}$ in V the sum $\mathbf{u} + \mathbf{v}$ is in V . (We say V is closed under addition.)
2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
4. There is a **zero** vector $\mathbf{0}$ in V such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$.
5. For each $\mathbf{u}$ in V , there is a vector $-\mathbf{u}$ in V such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.
6. For all $\mathbf{u}$ in V and all scalars c , the vector $c\mathbf{u}$ is in V . (We say V is closed under scalar multiplication.)
7. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$.
8. $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$.
9. $c(d\mathbf{u}) = (cd)\mathbf{u}$.
10. $1\mathbf{u} = \mathbf{u}$.

Examples:

1. The prototype examples of vector spaces are the $\mathbf{R}^n$ spaces that we have been working with in calculus and throughout this term so far.
2. Vector spaces of polynomials: The set $\mathbf{P}^n$ consists of all polynomials with real coefficients and degree at most n .

$$\mathbf{P}^2 =$$

$$\mathbf{P}^3 =$$

More examples of vector spaces

3.

4.

Definition: A **subspace** of a vector space V is a subset H of V that contains the zero vector, is closed under vector addition and closed under scalar multiplication.

Examples of Subspaces

1.

2.

Practice Proof

Let v_1, v_2, v_3 be any vectors in any vector space V . Let H be the $\text{span}\{v_1, v_2, v_3\}$.

Prove that H is a subspace of V .

Recall that the span of $\{v_1, v_2, v_3\}$ is _____

Step 1. The zero vector $\mathbf{0}$ is in H

Step 2. H is closed under addition.

Step 3. H is closed under scalar multiplication.